RoboCore Zambia

GET STARTED HERE

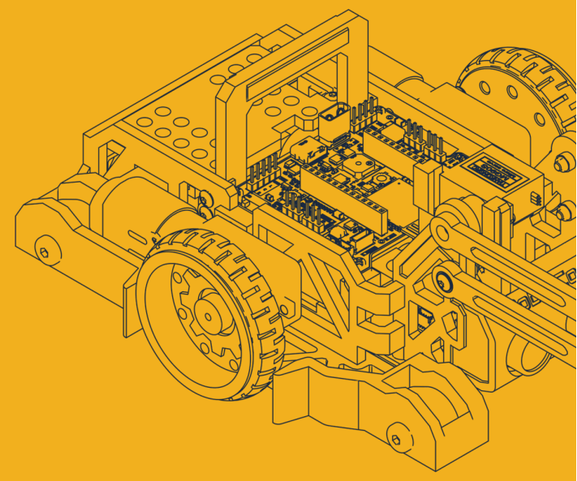
www.plastalbotbuilders.com

On this page you will find:

- **Assembly and first-run guide**
- **Step-by-step classroom lessons**
- **Programming environments: blocks, Python, C++**
- **Firmware, example code and libraries**
- **3D-print files for the frame, the lift and the gripper**
- **Support forum and teacher resources**

BEFORE YOU START

- 3 x 18650 lithium cells (not included) and a USB-C charger
- A computer or tablet with a browser, or the Arduino IDE
- Charge the robot fully before the first lesson
- Raise the lift by hand before you switch the robot on



Replacement parts and add-on sensors:
plastalbotbuilders.com/robocore

Design files are published. Print it, fix it, improve it, share it.

PLASTAL-BOT BUILDERS

Plastal-Bot Builders is a Lusaka makerspace that teaches robotics, coding and fabrication to learners and teachers across Zambia. RoboCore is our own controller family, designed so a class can build, program and repair its robots locally.

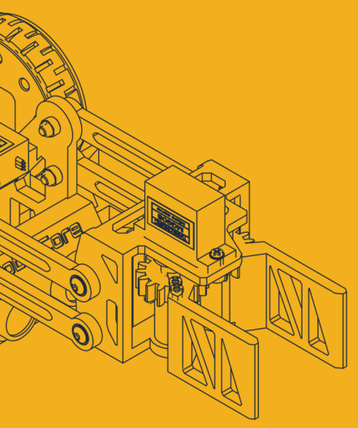
JOIN THE ROBOCORE COMMUNITY

Share what your class builds, find lesson plans and
schools. Replacement parts and add-on sensors a
design files are published so you can also print and

RoboCore box, inside, tile 1 of 6 (row 1, column 1), 1:1. Overlap 10 mm with the neighbouring tiles.

PLASTAL-BOT BUILDERS

SCAN FOR DOCS, CODE AND LESSONS



UNITY

lans and compare notes with other
ensors are available from us; the
print and fix parts yourself.

RoboCore box, inside, tile 2 of 2 (row 1, column 2). 211 × 211. Overlap 10 mm with the neighbouring tiles.

WORKSHOPS · TEACHER TRAINING · ROBOTICS CLUBS · FABRICATION

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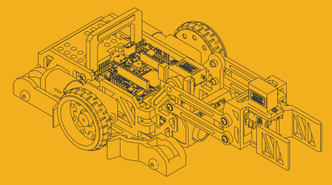
SCAN

LESSON PLANS · FORUM · SPARE PARTS · DESIGN FILES

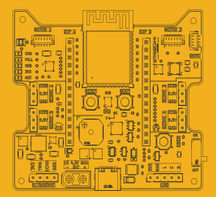
www.plastalbotbuilders.com/community

THIS KIT INCLUDES

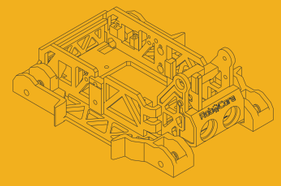
RoboCore robot, assembled



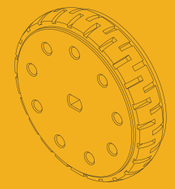
RoboCore V2 controller (on the robot)



3D-printed frame (on the robot)



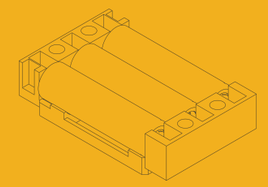
Wheels 56 mm, TPU tyres (2, on the robot)



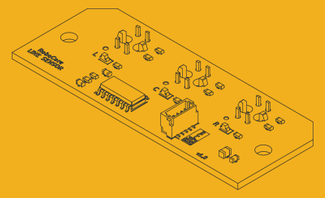
V-wheels 24 mm, with bearings (4, on the robot)



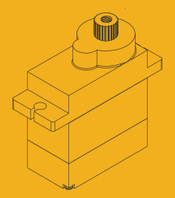
Battery cradle, 3 x 18650 (on the robot)



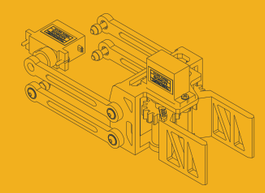
Line sensor, 3 channels (on the robot)



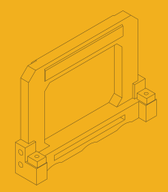
Micro servos MG90S (2, on the robot)



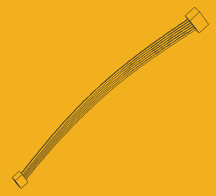
Lift and gripper (on the robot)



Carry handle (on the robot)



Motor encoder cables (2)



USB-C cable



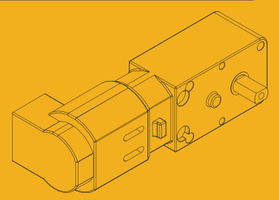
For replacement parts and add-on sensors visit plastalbotbuilders.com/robocore

RoboCore box, inside, tile 3 of 6 (row 2, column 1), 1:1. Overlap 10 mm with the neighbouring tiles.

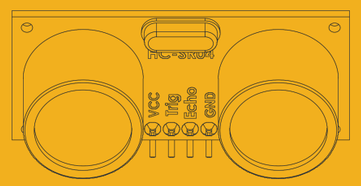
ILES

SCAN

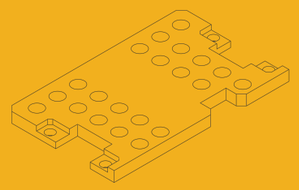
Encoder motors DG01D-E (2, on the robot)



Ultrasonic sensor HC-SR04 (on the robot)



Rear deck, Technic grid (on the robot)



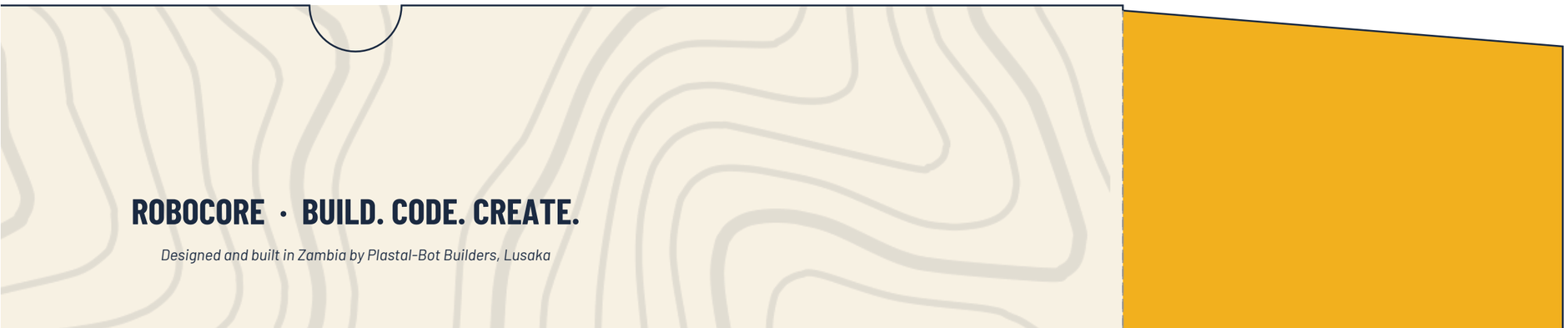
Screws and quick-start card



RoboCore box, inside, tile 4 of 6 (row 2, column 2), 1:1. Overlap 10 mm with the neighbouring tiles.

ROBOCORE · BUILD. CODE. CREATE.

Designed and built in Zambia by Plastal-Bot Builders, Lusaka



RoboCore box, inside, tile 5 of 6 (row 3, column 1), 1:1. Overlap 10 mm with the neighbouring tiles.



RoboCore box, inside, tile 6 of 6 (row 3, column 2), 1:1. Overlap 10 mm with the neighbouring tiles.